

Multiple Choice (10 marks)

1. Which of the following is/are NOT caused by orbital resonance?
 - (a) 2:3 periodic ratio of Neptune:Pluto
 - (b) Kirkwood Gaps.
 - (c) Gaps in Saturn's rings.
 - (d) Breaking of small Jovian moons to form ring materials.
2. Why can't we see the Sun's corona except during total solar eclipses?
 - (a) The corona is made up mostly of charged particles which aren't luminous.
 - (b) It's much too cool to emit visible light
 - (c) We can't see magnetic fields
 - (d) It's too diffuse
3. Meteorites with high metal content probably are
 - (a) pieces of comets rather than of asteroids.
 - (b) chunks of large differentiated asteroids that were shattered by collisions.
 - (c) chunks of rock chipped off the planet Mars.
 - (d) leftover chunks of rock from the earliest period in the formation of the solar system.
4. We were first able to accurately measure the diameter of Pluto from:
 - (a) a New Horizons flyby in the 1990 s
 - (b) Hubble Space Telescope images that resolved Pluto's disk
 - (c) brightness measurements made during mutual eclipses of Pluto and Charon
 - (d) radar observations made by the Arecibo telescope
5. Which of the following characteristics would not necessarily suggest that a rock we found is a meteorite.
 - (a) It has a fusion crust
 - (b) It contains solidified spherical droplets
 - (c) It is highly processed
 - (d) It has different elemental composition than earth

True / False (10 marks)

Answer True or False

6. True or False: A kilogram of bricks would weigh more than five kilograms of feathers on the Moon due to the density of the materials.
7. True or False: Knowing both the actual brightness and apparent brightness of an object allows you to estimate its distance from your location.
8. True or False: The axis of the Earth is tilted at an angle of approximately 23.5 degrees relative to its orbital plane around the Sun.
9. True or False: Cassiopeia is a prominent W-shaped constellation located in the northern sky, easily identifiable by its distinctive shape.
10. True or False: Dark energy is a theoretical model that explains the observed acceleration of the universe's expansion.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the effects on the spectral characteristics of a blackbody when its temperature is reduced to half of its original value, focusing on changes in power emitted and peak emission wavelength.
12. Discuss the significance of constellations in relation to the Milky Way, and analyze why the constellation Leo is not positioned along its plane.

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